

Wk 1 Notes

1. What kind of software & hardware are/will be on our drone? (Summary of the next 5 weeks)
 - a. Object Detection using OpenCV, switched to YOLO
 - b. Aerial Imagery/stitching w OpenCV, but OpenDroneMap has better results.
Offline Aerial Stitching
2. What kinds of challenges/projects can YOU work on as a Software team member?
 - a. Waypoint flying
 - b. Mapping/Surveying
 - c. Object detection, classification, localization (finding where physically an object is located)
 - d. Payload Delivery
 - e. Individual Projects
3. What's the state of modern drone autonomy software?
 - a. PX4 or ArduPilot for Flight Control (instead of Ardupilot/PX4/Betaflight)
 - b. ROS 2 (Robot Operating Sys 2) or custom for higher-lvl software
 - c. Autonomy: waypoint following → perception + obj localization + mission logic (ties to the software we'll be using)
 - d. Safety/Robustness: transition to redundant sensing, logging, etc
 - e. Important: BVLOS: Beyond Visual Line of Sight, and multiple drones collaborating/swarming
4. How does one learn about drones effectively in 2026?
 - a. AI lol
 - b. Other subteams